# Food waste prevention and redistribution

"we are going to develop an application to prevent surplus crop which is produced from a farmer and redirect it to a useful organisation like hotels,hostels etc and also redirect the food that becomes surplus in hotels and hostels etc to any other ngos and etc and we also use ai to prevent how food should we prepare in the hotels ,hostels etc after that if surplus exists we can redirect,THIS IS NOT ONLY ABOUT REDIRECTING FOOD IT ALSO CARES ABOUT PREVENTION OF FOOD" *Generated by LeapSpace*

## AI-Enabled Food Surplus Prevention and Redistribution at a Glance

### Key Findings Table

| Key finding | Relevant actors | Practical meaning for your app | Evidence |
|---|---|---|---|
| **Prevention-first works better than redistribution-only designs** | Hotels, hostels, caterers, kitchens, farms | Build forecasting, planning, and inventory controls as the core product; donation should be the second layer for unavoidable surplus only. | [1] [2] [3] [4] [5] [6] [7] [8] |
| **AI is strongest today in demand forecasting and waste tracking** | Kitchens, catering, HORECA, storage operations | Your most defensible AI modules are meal-demand prediction, batch planning, and waste measurement; end-to-end farm-to-NGO automation is promising but less proven. | [9] [10] [11] [12] [13] [14] [15] [16] |
| **Forecasting can materially reduce overproduction** | Catering, restaurants, food-service businesses | AI forecasts can support production planning and have been associated with reduced wasted meals, including reported reductions of **14% to 52%** in HoReCa forecasting deployments. | [7] [17] |
| **Waste tracking tools can also reduce waste after production begins** | HORECA, kitchens, buffet/catering settings | Add image/scanner/scale-based monitoring to identify where waste happens and continuously improve planning. | [13] [18] [19] [20] |

| Key finding | Relevant actors | Practical meaning for your app | Evidence |
|---|---|---|---|
| **Redistribution platforms need governance and trust, not just matching** | Donors, NGOs, volunteers, admins | Include recipient acceptance workflows, donor reliability records, scheduling discipline, and role-based coordination. | [21] [22] [23] [24] [25] [26] [27] |
| **Cold-chain and compliance are major bottlenecks for perishables** | NGOs, transporters, hotels, hostels, farms | Your app needs temperature, time-window, pickup, and traceability controls before enabling perishable redistribution. | [28] [29] [30] [31] [32] [33] [34] [35] |
| **Donation and resale have different economics** | Farmers, retailers, hospitality operators, NGOs | A mixed pathway—prevent, then resell/discount when viable, then donate—may be more sustainable than donation-only. | [36] [37] [38] [39] [40] [41] |
| **Evidence for fully integrated farm-to-hotel-to-NGO systems is still limited** | All stakeholders | Build the app modularly and measure outcomes carefully, because the literature is stronger on parts of the system than on complete end-to-end platforms. | [9] [11] [13] [14] [15] [16] |

### Direct Answer

Yes—your idea is well supported by the evidence **if the application is designed as a prevention-first system with redistribution as a fallback layer** rather than as a donation app alone [1] [2] [3] [4] [5] [7]. The strongest research-backed architecture is:

1. **Prevent farm and kitchen surplus before it is created** using AI forecasting, production planning, purchase alignment, occupancy/attendance signals, and waste tracking [7] [9] [10] [13].
2. **Route unavoidable surplus** from farmers to hotels/hostels and from hotels/hostels to NGOs using listing, matching, pickup scheduling, recipient acceptance, and delivery confirmation workflows [15] [21] [42] [43] [44] [45].
3. **Control safety and trust** with cold-chain checks, time limits, traceability records, and compliance features for perishable food [28] [29] [30] [46].

A strong MVP should therefore include: **farmer surplus listing, hotel/hostel demand forecasting, kitchen production recommendations, waste logging, surplus safety classification, NGO matching, pickup/rider coordination, recipient confirmation, and dashboards that separately measure "waste prevented" and "food redistributed"** [15] [21] [47] [48] [49].
**Confidence level: High** - This matches the clearest cross-source pattern and the most consistent implementation evidence [1] [9] [13] [21] [28].

### Study Scope

This report synthesizes evidence on:

- **Food-waste prevention** in hospitality, catering, and food service using forecasting, production planning, inventory alignment, and operational interventions [1] [2] [3] [4] [5] [6].
- **Surplus detection and redistribution** using apps, digital matching, AI/computer vision, scales, sensors, and community food-sharing systems [11] [12] [13] [14] [21].
- **Safe logistics and adoption conditions** including cold-chain monitoring, traceability, pickup scheduling, regulatory compliance, and organizational barriers [28] [29] [30] [46] [50] [51].

The underlying studies span roughly **2015-2025**, with many targeted reviews and pilots from **2018-2025** across hospitality, agri-food, cold-chain logistics, and digital platform systems [15] [52] [53].

### Assumptions & Limitations

- The supplied evidence **does not include strong peer-reviewed end-to-end evaluation** of one integrated application linking farmers, hotels/hostels, NGOs, and logistics with standardized outcome metrics such as pickup SLA, spoilage rate, and total net waste reduction [16] [54] [55] [56] [57].
- AI evidence is **much stronger for forecasting and tracking** than for fully automated surplus-routing decisions across multiple organizations [9] [10] [11] [12] [13] [14].
- Donation-system papers often report feasibility or satisfaction, but **many omit comparable operational metrics**, which limits benchmarking [3] [18] [54] [55] [56] [57].
- Perishable-food redistribution is constrained by **cold-chain, hygiene, and traceability** requirements that may make some surplus unsuitable for donation even if matching demand exists [28] [31] [32] [33].

### Suggested Further Research

- Run a pilot that measures **both prevention and redistribution KPIs** separately: forecast error, kilograms prevented, kilograms donated, pickup completion, spoilage rate, and cost per kilogram recovered [15] [48] [49].
- Test **hotel/hostel-specific forecasting features** such as occupancy, reservations, holidays, weather, and meal-type patterns, because sector-specific features remain underused [58] [59] [60] [61].

- Evaluate a **safe donation window score** using preparation/harvest time, temperature history, packaging state, and travel time; the literature supports each component separately but not yet as one deployed scoring system [46] [62] [63] [64].
- Compare **donation-only versus mixed resale-plus-donation models** for different shelf-life windows [38] [39] [40] [41].
- Study **farmer-to-hotel surplus matching** directly, because the evidence base is thinner there than for restaurant/retail-to-NGO redistribution [15] [47].

## Full Report

## Foundational Context

Your proposed application combines **two different but connected problems**:

1. **Prevention**: stopping surplus before it becomes waste by predicting real demand and adjusting production, procurement, and harvest allocation [1] [2] [3] [4] [5].
2. **Redistribution**: moving unavoidable edible surplus to another user fast enough and safely enough to preserve value [21] [28] [42] [43] [46].

The evidence consistently places these in a **waste hierarchy**: first prevent overproduction, then recover edible surplus through donation or other channels, rather than treating donation as the main solution [1] [2] [3] [4] [5] [6] [8]. This matters because prevention attacks the largest avoidable losses—wrong forecasting, over-preparation, poor inventory alignment, and buffet/catering overproduction—while redistribution only works for the portion that remains edible, safe, and logistically movable [28] [65] [66] [67].

Taken together, these findings suggest your application should be designed as a **multi-layer decision system**:

- predict demand and production need,
- detect emerging surplus,
- classify what is still usable,
- route it to the best recipient or recovery path [11] [13] [14] [46] [68] [69].

**Confidence level: High** - This framing is directly supported across prevention, redistribution, and logistics evidence [1] [21] [28].

# 1) Preventing Food Surplus Before It Is Created

### 1.1 Demand forecasting for kitchens and meal services

The strongest evidence for your app is in **AI-based demand forecasting**. Across food-service settings, machine-learning models such as Random Forest, XGBoost, LSTM, boosting methods, and ensembles are repeatedly used to predict demand or wastage and to improve production and inventory planning [1] [2] [3] [4] [5] [9] [70]. Restaurant and menu-level systems use historical sales and contextual factors to forecast item demand, which helps reduce overproduction and support better stock ordering [9] [10].

The most concrete operational result in the supplied evidence is that **HoReCa forecasting deployments reduced wasted meals by 14% to 52%** by helping businesses decide how much to produce in advance [7]. Short-term demand forecasting in catering was also explicitly framed as a tool to reduce overproduction and food waste by aligning production with expected demand [17].

Comparative model evidence exists, but it is still patchy for hotels and hostels specifically. A 2025 restaurant food-supply-chain study compared **LSTM, XGBoost, and Random Forest**, with **XGBoost performing best** on residual accuracy, waste-to-sales ratio, and cost savings, although the abstract did not report exact percentage outcomes [3]. A 2023 cafeteria-meal study found **CatBoost** best, and a stacking ensemble improved performance by **8%**, but explicit waste-reduction figures were not given [71]. A 2025 breakfast-sales study found **Random Forest** most accurate among tested models for one business context, again without direct waste-reduction outcomes [72]. No retrieved paper explicitly provided the user-requested **hotel/hostel-specific RF vs XGBoost vs LSTM comparison with downstream waste outcomes** [3] [71] [72].

For your application, that means the safest implementation choice is not to assume one universal best model. Instead, the evidence supports building a forecasting layer that can test multiple models and choose the best one for each deployment context [3] [71] [72].

#### Practical implication for the app

- Forecast number of meals by location, date, meal type, and menu item [9] [10].
- Convert forecast into suggested preparation quantity and purchase quantity [2] [3] [4] [5] [73] [74].
- Track forecast error and adjust model choice over time [3] [71].

**Confidence level: High** - Forecasting for prevention is one of the most repeatedly supported themes, though hotel/hostel-specific comparative evidence is thinner [1] [7] [9].

### 1.2 Menu, purchase, and batch-size optimization

Forecasts are only useful if they change operations. Several studies directly frame ML as a way to optimize **purchase decisions, stock levels, and inventory responsiveness**, thereby reducing overstocking and spoilage in restaurants and meal-supply businesses [2] [3] [4] [5] [70] [73] [74]. In hospitality, predictive analytics is linked not just to cost efficiency but also to food-waste reduction and operational support for zero-waste planning [6] [75] [76].

This is important for hotels and hostels because surplus often emerges not only from bad demand estimates but also from **fixed batch habits**, overly conservative purchasing, and buffet-style provisioning [65] [66] [67]. Evidence from hospitality waste studies shows avoidable waste occurs in kitchens, serving, and plate waste, especially in buffet settings and overproduction contexts [65] [66] [67].

A useful design inference is that your app should not stop at "predicted demand = 300 meals." It should translate that into:

- recommended purchase quantities,
- recommended batch sizes,
- staggered cooking schedules,
- alert thresholds for overproduction risk [2] [4] [5] [68] [73] [74].

Taken together, the literature suggests that **production planning is the mechanism through which forecasting becomes waste prevention** [7] [17] [68].

**Confidence level: High** - The prevention logic is strongly supported, even though explicit optimization-model details are less standardized across papers [2] [5] [7] [73].

### 1.3 Context-aware prediction features

Forecast accuracy improves when models include **context**, not just historical averages [70] [77] [78]. The retrieved evidence highlights several useful feature classes:

- **Calendar and temporal signals** such as day-of-week, hour-of-day, holidays, and repeated seasonal periods [52] [60] [79].
- **Promotions and campaign effects**, which can substantially change demand [79].
- **Weather and exogenous conditions**, which influence both dining and delivery demand [59] [80].
- **Occupancy and hotel-demand proxies**, including review sentiment/topic data and related exogenous variables in hotel forecasting work [58] [59].

However, the evidence also shows a gap: there is **limited explicit use of ingredient-level history and hostel/hotel-specific meal-service variables** in the retrieved set [52] [61]. This creates a practical opportunity for your system.

For hotels and hostels, especially institutional kitchens, potentially high-value features include:

- booking counts,
- room occupancy,
- attendance/leave records,
- exam periods or holidays for hostels,
- special events,
- menu attractiveness,
- prior leftovers by dish [13] [58] [59] [60].

This last step is partly an inference from the literature's emphasis on contextual improvement. **Inferred:** richer institutional features are likely to improve forecast usefulness for hostels and canteens, though this specific combination was not directly benchmarked in the supplied studies [52] [59] [61]. **Confidence level: Medium** - The importance of context is well supported, but some proposed features are extrapolated to your use case [58] [60].

## 2) Detecting and Redistributing Surplus Food

### 2.1 Surplus detection and waste measurement

Once production begins, a second AI layer becomes useful: **waste detection and measurement**. The evidence shows that AI, computer vision, sensors, and scales can be used to detect leftovers, spoilage, disposal anomalies, and waste quantity in real time [11] [12] [13] [14] [19].

One HORECA intervention using AI-based waste tracking reported **food-waste reductions of 23% to 51%** and reduced wasted-food cost per meal by up to **39%** [13]. This is important because it shows that even after prevention planning, real-time tracking can produce additional savings and feedback for learning.

Examples in the retrieved evidence include:

- tray-image leftover recognition with automated estimation, where one system reported "excellent" results for **39% of dishes** and "good" results for **19%** [20];
- scale-and-tablet buffet leftover reporting directly at the source [18];
- digitized systems detecting anomalies and linking meal reservations/attendance with production adjustment [19].

For your app, this supports a **closed loop**:

1. predict demand,
2. recommend preparation volume,

3. record actual production,
4. measure leftovers and spoilage,
5. retrain future recommendations [13] [14] [19].

At the farm and storage stage, AI/ML and IoT interventions can also help reduce post-harvest losses and improve cold storage and transport decisions, though demand prediction and overproduction remain harder challenges there [14] [81] [82].

**Confidence level: High** - Waste detection is well supported for monitoring and improvement, though exact accuracy metrics vary by pilot [13] [20].

### 2.2 App-based matching and donation workflows

For redistribution, the evidence strongly supports **digital matching platforms** that connect food providers and recipients via dashboards, location-based matching, administrative coordination, and real-time visibility of available items [11] [21] [42] [43] [44] [45].

The clearest quantified pilot in the supplied set is a **2022 mobile-app redistribution pilot** that, over 11 months, engaged **50 food businesses**, recovered **43,900 pounds** of food, delivered to **34 community sites**, and served **28,400 meals** [15]. Other platforms connected hotels, NGOs, restaurants, hostels, and volunteers, and often included mapping, visibility, or acceptance workflows, but many did not report hard matching or pickup metrics [54] [55] [56] [57].

This uneven evidence implies that your app should be designed not only to function operationally but also to **measure the metrics missing from prior pilots**:

- listing-to-match rate,
- recipient acceptance rate,
- pickup completion rate,
- time from listing to pickup,
- spoilage before pickup,
- successful delivery rate [54] [55] [56] [57].

Recommended workflow:

1. donor lists surplus batch,
2. system classifies food type and time sensitivity,
3. platform proposes nearby recipients,
4. recipient accepts or declines,
5. rider/volunteer pickup scheduled,

6. delivery confirmed,
7. chain-of-custody logged [15] [21] [43] [45] [46].

**Confidence level: High** - Platform workflows are well evidenced; standardized performance reporting is the main weakness in the literature [15] [21].

### 2.3 Redistribution models: donation, resale, or mixed pathways

The evidence does **not** support a single redistribution model as universally best. Instead, it shows trade-offs.

- **Direct NGO donation** generates social value and environmental benefit, but often requires public or private support and may impose net financial losses on operators [39] [40].
- A comparative Danish study found donation could compete with discard or nonfood uses, but **not with sale on alternative platforms** when expiry timing still allowed commercial sale [38].
- **Mixed community models** involving municipalities, retailers, and aid organizations are more flexible than traditional food-bank-only models [15] [83].
- **Resale and social-supermarket models** can preserve food value while also improving user choice and autonomy [41] [84].
- Digital marketplace models increasingly support both donation and surplus purchasing [24] [47] [85].

For your application, this suggests a routing engine that follows a hierarchy:

1. prevent,
2. use internally or transfer to another buyer,
3. discount/resell if allowed and viable,
4. donate if safe and timely,
5. divert to non-human recovery if needed.

This routing logic is partly inferential but directly grounded in the evidence on economic trade-offs and model diversity [38] [39] [40] [41].
**Confidence level: High** - The trade-off between social value and economic viability is explicit across sources [36] [37] [38].

## 3) Safe Logistics, Governance, and Adoption

### 3.1 Cold chain, pickup, and traceability

Perishable-food redistribution succeeds only if logistics preserve safety. The evidence consistently identifies **refrigerated transport limits, temperature-control failures, weak last-mile cold-chain capacity, coordination problems, and traceability gaps** as major bottlenecks [28] [29] [30] [31] [32] [33] [34] [35].

The most frequently reported operational problems include:

- breaks in cold-chain integrity [31] [33],
- weak infrastructure and last-mile cold-chain gaps [32] [34],
- scheduling and transparency issues [86] [87],
- high cost and monitoring limitations [88] [89] [90].

For your app, this means each surplus batch should carry at least:

- timestamp,
- origin,
- quantity,
- storage condition,
- preparation/harvest time,
- expiry estimate,
- pickup deadline,
- transport requirement [46] [62] [63].

A practical platform rule would be: **the system should refuse or restrict matching if the safe pickup window or storage requirements cannot be met**. This is a direct operational implication of the cold-chain evidence [28] [31] [33].

**Confidence level: High** - Cold-chain constraints are among the strongest and most consistent findings [28] [32] [35].

### 3.2 Food safety, compliance, and trust mechanisms

Traceability technologies such as **blockchain, IoT, and AI** are presented as ways to improve food safety, quality monitoring, tamper resistance, and compliance in food logistics [29] [30]. The evidence does **not** provide a full peer-reviewed donated-food traceability evaluation covering freshness verification, tamper resistance, compliance, and delivery speed together in one real-world NGO system [16]. Still, adjacent evidence supports each function separately:

- **Freshness verification**: IoT/AI can capture real-time temperature, humidity, location, and condition data for quality and shelf-life assessment [62] [63] [64].
- **Tamper resistance**: blockchain-based records are presented as immutable and decentralized [91] [92] [93] [94].
- **Regulatory compliance**: smart contracts and digital records can support verification and automate compliance steps [95] [96] [97].
- **Faster response and traceability**: blockchain and IoT systems can speed tracing and support quicker decisions [16] [63] [93] [97].

A sensible application design is therefore a lightweight **surplus passport** for each batch: origin, batch ID, allergen info, timestamps, storage record, recipient, and delivery confirmation [46] [62] [94]. **Confidence level: Medium** - Individual technical benefits are supported, but donated-food-specific integrated validation remains limited [16].

### 3.3 Governance, incentives, and long-term adoption

Technology alone is not enough. Redistribution platforms depend on **trust, governance, donor-recipient relationships, timing discipline, and business-model fit** [22] [23] [24] [25] [26] [27]. FareShare-style systems, for example, operate through brokered governance structures, but donor-store-manager relationships and logistical arrangements strongly affect continued participation [25]. Studies also note that actors must distinguish **genuine waste from redistributable surplus**, otherwise utilization remains partial [26].

On the implementation side, recurring barriers include:

- low interoperability and siloed data [98],
- low willingness to share data [98],
- skills gaps and training needs [99],
- regulatory and policy challenges [50] [51] [100],
- cost and resource constraints [101] [102],
- privacy and cybersecurity concerns [103].

Enablers include stronger government backing, infrastructure support, standards, clearer coordination roles, and more professionalized operational processes [28] [69] [100] [104].

For your app, that means governance features are not optional. You likely need:

- verified organization accounts,
- role-based permissions,
- recipient acceptance rules,

- audit logs,
- cancellation and no-show handling,
- data-sharing consent controls [27] [98] [103].

**Confidence level: High** - Governance and adoption barriers are strongly and repeatedly supported [25] [98] [100].

## 4) Recommended Product Architecture for Your Application

Based on the evidence, the strongest application architecture is a **modular prevention-and-redistribution platform**.

### Core modules

#### A. Farmer surplus intake

Farmers list available surplus crop, quantity, condition, expected harvest date, and time sensitivity; the platform then matches this to hotels, hostels, or other bulk buyers where appropriate [21] [43] [45]. Evidence for farmer-specific matching is thinner than for food-service donation, so this module should be piloted carefully [15] [47].

#### B. Hotel/hostel demand forecasting

Predict meal demand using historical consumption and context such as occupancy, date, holidays, and weather [9] [10] [58] [59] [60].

#### C. Production recommendation engine

Translate forecasts into suggested purchasing and staged batch production [2] [4] [5] [73] [74].

#### D. Waste tracking

Use manual logging at minimum; optionally add image/scanner/scale-based measurement for leftovers and buffet waste [13] [18] [20].

#### E. Surplus classification

Classify surplus by type, condition, shelf-life, and transport needs before routing [28] [46] [62].

#### F. Matching and routing

Match surplus to NGOs, shelters, community kitchens, or other recipients; include pickup windows, volunteer/rider assignment, and recipient confirmation [15] [21] [48] [105] [106].

#### G. Traceability and compliance

Store chain-of-custody, delivery confirmation, and condition records [29] [30] [46].

### H. Analytics dashboard

Separate **prevention KPIs** from **redistribution KPIs**, because the literature treats them as different outcomes [15] [49].

**Confidence level: High** - This architecture closely reflects the strongest evidence-backed functions, though end-to-end validation remains a research gap [9] [13] [15] [46].

## 5) MVP and KPI Recommendations

### MVP features

For a first release, the evidence most strongly supports:

- donor/recipient registration and verification [21] [27];
- farmer and hotel surplus listing [21] [43];
- hotel/hostel meal-demand forecasting [9] [10];
- production quantity recommendations [2] [4] [73];
- surplus safety/time-window tagging [28] [46];
- NGO matching and acceptance workflow [15] [57];
- pickup scheduling and confirmation [48];
- waste/prevention dashboard [13] [49].

### KPIs

The literature implies two headline KPI families:

### Prevention KPIs

- forecast accuracy [3] [71];
- meals overproduced vs target [7];
- kilograms prevented from becoming surplus [13] [49];
- purchase alignment / inventory overstock reduction [2] [5] [73].

### Redistribution KPIs

- kilograms listed and matched [15];
- recipient acceptance rate [57];
- pickup completion rate [48];
- time from listing to pickup [46];

- spoilage before delivery [31] [33];
- meals served from recovered food [15].

**Confidence level: High** - These KPIs directly address the metric gaps identified in the evidence [54] [56].

## 6) Final Assessment

The research strongly supports your concept **when framed as an AI-enabled food-surplus prevention system with redistribution as a controlled secondary pathway** [1] [2] [3] [4] [5] [21] [28]. The best-supported parts are:

- AI demand forecasting for kitchens [7] [9],
- operational planning and inventory alignment [2] [5] [73],
- waste tracking and detection [13] [14],
- app-based donor-recipient coordination [15] [21],
- cold-chain and traceability safeguards [28] [46].

The weakest evidence area is a **fully integrated, peer-reviewed farm-to-hotel-to-NGO platform with standardized end-to-end outcomes** [16] [54] [55]. That is not a reason to avoid the idea; rather, it means your project can create value by implementing the system modularly and measuring it rigorously [15] [48].

**Bottom line:** build the app around **prevent -> detect -> classify -> route -> verify -> learn**, with prevention as the primary product promise and redistribution as the fallback for unavoidable surplus [11] [13] [46] [68].
**Overall confidence level: High** - The strategic direction is strongly supported across the supplied evidence, though some integrated workflow details remain under-evaluated in the literature [1] [21] [28].

## References

1. Kumar, J.D.S., Nayudu, P.P., M, M.N., (...), Parimala, V. (2026) Predictive Modeling of Food Demand using Machine Learning - A Solution to Reduce Food Waste. *Proceedings of 4th International Conference on Electronics and Renewable Systems, ICEARS 2026*. https://doi.org/10.1109/ICEARS67481.2026.11416566
2. Mugdha, Sharma, D., Jose, D. (2024) Transforming Inventory Management Through Machine Learning based Demand Forecasting for a Circular Economy. *3rd International Conference on Communication, Control, and Intelligent Systems, CCIS 2024*. https://doi.org/10.1109/CCIS63231.2024.10932031

3. Chowdhury, S., Prasad, P., Patidar, L., (...), Sharma, M. (2025) Predictive Analytics in Food Supply Chain and Waste Management. *2025 5th International Conference on Advancement in Electronics and Communication Engineering, AECE 2025*. https://doi.org/10.1109/AECE67531.2025.11386707
4. Olivos, M., Motta, A., Castañeda, P. (2025) Optimization of Food Inputs in Restaurants in Metropolitan Lima Through Prediction and Monitoring Based on Machine Learning. *International Conference on Cloud Computing and Services Science, CLOSER - Proceedings*. https://doi.org/10.5220/0013233700003950
5. Phatangare, S., Sharma, L., Mate, A., (...), Pardeshi, D. (2025) Revolutionizing Food Ordering: An ML-Driven System for Predictive Sales and Inventory Optimization. *2025 International Conference on Computing Technologies, ICOCT 2025*. https://doi.org/10.1109/ICOCT64433.2025.11118931
6. Awad, N.A., Juneja, S., Mapari, S., (...), Mahmoud, A. (2025) The Impact of Employee's Awareness of Food Safety in Egyptian Hotels: Machine Learning Approach. *Procedia Computer Science*. https://doi.org/10.1016/j.procs.2025.03.319
7. Seyam, A., Samuel Mathew, S.S., Barachi, M.E., (...), Shen, J. (2026) The application of machine learning and deep learning on demand forecasting across time-critical industries: A systematic review. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2026.104625
8. Kallbekken, S., Sælen, H. (2013) 'Nudging' hotel guests to reduce food waste as a win-win environmental measure. *Economics Letters*. https://doi.org/10.1016/j.econlet.2013.03.019
9. Acı, M., Yergök, D. (2023) Demand Forecasting for Food Production Using Machine Learning Algorithms: A Case Study of University Refectory. *Tehnicki Vjesnik*. https://doi.org/10.17559/TV-20230117000232
10. Kim, S. (2025) Development of an AI-based restaurant menu demand prediction model utilizing sales and meteorological data. *Food Science and Biotechnology*. https://doi.org/10.1007/s10068-025-01956-2
11. Poongodi, K., Sathish Kumar, M., Srinithi, R., (...), Mohana Swara Raj, N. (2026) Bridging the Gap Between Donors and Recipients: A Digital Application to Food Donation. *Proceedings of 5th International Conference on Communication, Computing and Electronics Systems, ICCCES 2026*. https://doi.org/10.1109/ICCCES62661.2026.11436954
12. Manikanta, A.K., Malarvizhi, N. (2025) AI-Powered Food Waste Reduction System. *Proceedings - IEEE 10th International Conference on Smart Structures and Systems, ICSSS 2025*. https://doi.org/10.1109/ICSSS66939.2025.11346403
13. Sigala, E.G., Gerwin, P., Chroni, C., (...), Lasaridi, K. (2025) Reducing food waste in the HORECA sector using AI-based waste-tracking devices. *Waste Management*. https://doi.org/10.1016/j.wasman.2025.02.044
14. Onyeaka, H., Akinsemolu, A., Miri, T., (...), Ugwa, C. (2025) Artificial intelligence in food system: Innovative approach to minimizing food spoilage and food waste. *Journal of Agriculture and Food Research*. https://doi.org/10.1016/j.jafr.2025.101895

15. Ayala, V., Caldwell, J.I., Garcia-Silva, B., (...), Kuo, T. (2022) Increasing Surplus Food Redistribution to Improve Food Access Through a Partnership Between Public Health and a Technology-Based Company. *Journal of Health Care for the Poor and Underserved*. https://doi.org/10.1353/hpu.2022.0156

16. Ellahi, R.M., Wood, L.C., Ahmed Bekhit, A.E.-D.A. (2024) Blockchain-Driven Food Supply Chains: A Systematic Review for Unexplored Opportunities. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app14198944

17. Rodrigues, M., Miguéis, V., Freitas, S., Machado, T. (2024) Machine learning models for short-term demand forecasting in food catering services: A solution to reduce food waste. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2023.140265

18. Leverenz, D., Moussawel, S., Hafner, G., Kranert, M. (2020) What influences buffet leftovers at event caterings? A German case study. *Waste Management*. https://doi.org/10.1016/j.wasman.2020.07.029

19. Cantarelli da Silva, Regiana, Bacharini Lima, Lívia, Batistela dos Santos, Emanuele, Akutsu, Rita de Cássia (2025) The Use of Industry 4.0 and 5.0 Technologies in the Transformation of Food Services: An Integrative Review. *Foods*. https://doi.org/10.3390/foods14244320

20. Sari, Yuita Arum, Nakazawa, Atsushi, Wani, Yudi Arimba (2025) LeFood-set: Baseline performance of predicting level of leftovers food dataset in a hospital using MT learning. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0320426

21. Sudha, D.K.S.N., Rajamohan, N., Priya, C.H., (...), Ramasamy, G. (2026) Food Donation and Redistribution System Using Database. *Advanced Transportation Systems in the Field of Engineering and Technology: Volume 2*. https://doi.org/10.1201/9781042005376-98

22. de Almeida Oroski, F., da Silva, J.M. (2023) Understanding food waste-reducing platforms: A mini-review. *Waste Management and Research*. https://doi.org/10.1177/0734242X221135248

23. Cane, M., Parra Rodríguez, C. (2020) Digital platforms: mapping the territory of new technologies to fight food waste. *British Food Journal*. https://doi.org/10.1108/BFJ-06-2019-0391

24. Yu, M., Secondi, L., Laureti, T., Palumbo, L. (2025) Saving food surplus and developing new business models: Exploring the potential of 'Too Good To Go' at territorial level using web-scraped data. *Big Data Research*. https://doi.org/10.1016/j.bdr.2025.100536

25. Alexander, C., Smaje, C. (2008) Surplus retail food redistribution: An analysis of a third sector model. *Resources, Conservation and Recycling*. https://doi.org/10.1016/j.resconrec.2008.07.009

26. Midgley, J.L. (2014) The logics of surplus food redistribution. *Journal of Environmental Planning and Management*. https://doi.org/10.1080/09640568.2013.848192

27. Schneider, F. (2013) The evolution of food donation with respect to waste prevention. *Waste Management*. https://doi.org/10.1016/j.wasman.2012.10.025

28. de Boeck, E., Jacxsens, L., Goubert, H., Uyttendaele, M. (2017) Ensuring food safety in food donations: Case study of the Belgian donation/acceptation chain. *Food Research International*. https://doi.org/10.1016/j.foodres.2017.08.046

29. Radhika, K., Mohammed, S., Tadepalli, S.K., Puvulla, K. (2024) Food supply chain management by leveraging AI, IoT, and blockchain technologies. *The Future of Agriculture: IoT, AI and Blockchain Technology for Sustainable Farming*. https://doi.org/10.2174/9789815274349124010010

30. S, S., B K, B., Debnath, S., Mudliar, S.N. (2023) Advanced Computational Tools for Enhanced Food Quality and Safety. *Food Engineering Series*. https://doi.org/10.1007/978-3-031-30683-9_8

31. Cheng, H., Zhou, X., Sun, J., (...), Liu, B. (2024) Assessing the efficacy of agricultural cold chain facility expansion in China. *Journal of Stored Products Research*. https://doi.org/10.1016/j.jspr.2024.102244

32. Han, J.-W., Zuo, M., Zhu, W.-Y., (...), Yang, X.-T. (2021) A comprehensive review of cold chain logistics for fresh agricultural products: Current status, challenges, and future trends. *Trends in Food Science and Technology*. https://doi.org/10.1016/j.tifs.2021.01.066

33. Chen, C., Gao, W.-W., Hu, Z.-H., (...), Pan, S.-D. (2019) Assigning customer-dependent travel time limits to routes in a cold-chain inventory routing problem. *Computers and Industrial Engineering*. https://doi.org/10.1016/j.cie.2019.05.018

34. Nautiyal, M., Joshi, S., Hussain, I., (...), Kumar, S. (2025) Revolutionizing agriculture: A comprehensive review on artificial intelligence applications in enhancing properties of agricultural produce. *Food Chemistry: X*. https://doi.org/10.1016/j.fochx.2025.102748

35. Zanoni, Simone, Marchi, Beatrice (2021) Environmental impacts of foods refrigeration. *Environmental Impact of Agro-Food Industry and Food Consumption*. https://doi.org/10.1016/B978-0-12-821363-6.00007-2

36. Maleki Vishkaei, B., De Giovanni, P. (2026) Smart food-sharing platforms for social sustainability: a heuristic algorithm approach. *International Transactions in Operational Research*. https://doi.org/10.1111/itor.13543

37. Onyeaka, H., Tamasiga, P., Nwauzoma, U.M., (...), Akinsemolu, A.A. (2023) Using Artificial Intelligence to Tackle Food Waste and Enhance the Circular Economy: Maximising Resource Efficiency and Minimising Environmental Impact: A Review. *Sustainability (Switzerland)*. https://doi.org/10.3390/su151310482

38. Jensen, J.D. (2025) Food Waste Prevention and Economic Incentives to Redistribute Surplus Foods from Food Retailing. *Journal of Food Products Marketing*. https://doi.org/10.1080/10454446.2025.2584844

39. Sundin, N., Bartek, L., Persson-Osowski, C., (...), Eriksson, M. (2023) Sustainability assessment of surplus food donation: A transfer system generating environmental, economic, and social values. *Sustainable Production and Consumption*. https://doi.org/10.1016/j.spc.2023.03.022

40. Bergström, P., Malefors, C., Strid, I., (...), Eriksson, M. (2020) Sustainability assessment of food redistribution initiatives in Sweden. *Resources*. https://doi.org/10.3390/resources9030027

41. Payling, R., Davies, A., Zimmermann, S., Downey, C. (2025) A pre and post analysis of food and carbon flows of a surplus food café initiative. *Proceedings of Institution of Civil Engineers: Waste and Resource Management*. https://doi.org/10.1680/jwarm.24.00010

42. Jakhmola-Mani, R.J., Singh, S., Sonu, S., Pande Katare, D.P. (2025) Tech-Driven Harvest: Blockchain's Role in Sustainable Food Systems and Waste-Free Future. *Blockchain and Digital Twin Applications in Smart Agriculture*. https://doi.org/10.1201/9781003507390-13

43. Chandana, S., Sooda, G., Patil, A., Manonmani, S. (2025) FeedBridge: An AI-Powered Platform for Sustainable Food Redistribution. *Proceedings of IEEE International Conference for Women in Innovation, Technology and Entrepreneurship, ICWITE 2025*. https://doi.org/10.1109/ICWITE64848.2025.11307073

44. Abitha, S., Lathika Leyandra, M., Varshidha, M., (...), Sangeetha, M. (2026) FoodResQ - Food Donation Website. *2026 International Conference on Innovations in Emerging Technologies for Sustainable Development: Empowering Innovation for a Sustainable Future, ICIETSD 2026 - Conference Proceedings*. https://doi.org/10.1109/ICIETSD68684.2026.11584660

45. Kawengian, D.J., Hizkia, N.N., Taju, S.W. (2024) Android-Based Restaurant Food Donation and Distribution App for Helping Hands Using Global Positioning System. *2024 6th International Conference on Cybernetics and Intelligent System, ICORIS 2024*. https://doi.org/10.1109/ICORIS63540.2024.10903796

46. Gnanabaskaran, A., Sakthi, R., Nithesh, T. (2026) Donato: An AI-Driven Food Donation and Redistribution Platform for Reducing Waste and Hunger. *Proceedings of 2nd International Conference on Multi-Agent Systems for Collaborative Intelligence, ICMSCI 2026*. https://doi.org/10.1109/ICMSCI67830.2026.11469451

47. Eriksson, N., Stenius, M. (2024) Consumer Intention to Use Mobile Applications for Buying Surplus Food: A Research Model. *IEEE International Conference on Industrial Engineering and Engineering Management*. https://doi.org/10.1109/IEEM62345.2024.10857223

48. Davis, L.B., Sengul, I., Ivy, J.S., (...), Miles, L. (2014) Scheduling food bank collections and deliveries to ensure food safety and improve access. *Socio-Economic Planning Sciences*. https://doi.org/10.1016/j.seps.2014.04.001

49. Pirani, S.I., Arafat, H.A. (2016) Reduction of food waste generation in the hospitality industry. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2015.07.146

50. Kumar, R., Samadhiya, A., Kumar, A., (...), El Jaouhari, A. (2025) Implementing AI in agri-food supply chain security from the multi-stakeholder perspective: an exploratory review and future directions. *International Journal of Industrial Engineering and Operations Management*. https://doi.org/10.1108/IJIEOM-01-2025-0011

51. Korcari, A., Saridi, M., Koumpoti, A., Anastasiadis, F. (2026) AI-Driven Demand Planning: A Systematic Review of Adoption, Barriers and Strategic Implications. *Administrative Sciences*. https://doi.org/10.3390/admsci16060260

52. Posch, K., Truden, C., Hungerländer, P., Pilz, J. (2022) A Bayesian approach for predicting food and beverage sales in staff canteens and restaurants. *International Journal of Forecasting*. https://doi.org/10.1016/j.ijforecast.2021.06.001

53. Theophilus, O., Dulebenets, M.A., Pasha, J., (...), Mazaheri, A. (2021) Truck scheduling optimization at a cold-chain cross-docking terminal with product perishability considerations. *Computers and Industrial Engineering*. https://doi.org/10.1016/j.cie.2021.107240

54. Nandeshwar, V., Talathi, S.S., Sarangdhar, R.S., (...), Sarode, V.K. (2025) Ending hunger, saving food: Innovations for surplus redistribution for NGO's. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0291696

55. Chitrala, C., Makka, S., Reddy, S.B., (...), Budati, B.A. (2025) Smart food donation: An Android app for efficient waste management. *Security Issues in Communication Devices, Networks and Computing Models: Volume 2*. https://doi.org/10.1201/9781003591788-14

56. Alsayaydeh, J.A.J., Bacarra, R., Abd Gani, S.F.B., (...), Herawan, S.G. (2025) FoodSharePro: An Integrated Mobile Platform for Sustainable Food Donation and Decentralized Composting. *International Journal of Advanced Computer Science and Applications*. https://doi.org/10.14569/IJACSA.2025.0160711

57. Haseena Barveen, H., Durga, P., Baviya Shree, B., Patturose, G.B. (2025) FoodBridge: Connecting Surplus to Need. *2025 International Conference on Emerging Technologies in Engineering Applications, ICETEA 2025 - Proceedings*. https://doi.org/10.1109/ICETEA64585.2025.11099979

58. Wu, D.C., Zhong, S., Song, H., Wu, J. (2024) Do topic and sentiment matter? Predictive power of online reviews for hotel demand forecasting. *International Journal of Hospitality Management*. https://doi.org/10.1016/j.ijhm.2024.103750

59. Huang, L., Zheng, W. (2021) Novel deep learning approach for forecasting daily hotel demand with agglomeration effect. *International Journal of Hospitality Management*. https://doi.org/10.1016/j.ijhm.2021.103038

60. https://www.sciencedirect.com/science/article/pii/S2948136825000025

61. Lasek, A., Cercone, N., Saunders, J. (2016) Smart restaurants: Survey on customer demand and sales forecasting. *Smart Cities and Homes: Key Enabling Technologies*. https://doi.org/10.1016/B978-0-12-803454-5.00017-1

62. Kaur, A., Singh, G., Kukreja, V., (...), Yoon, B. (2022) Adaptation of IoT with Blockchain in Food Supply Chain Management: An Analysis-Based Review in Development, Benefits and Potential Applications. *Sensors*. https://doi.org/10.3390/s22218174

63. Sunkara, S.K. (2025) LEVERAGING AI, IoT, AND BLOCKCHAIN FOR SCALABLE DIGITAL TRANSFORMATION IN POST-HARVEST SUPPLY CHAINS: A MULTI-SECTOR APPROACH TO ENHANCING EFFICIENCY AND TRACEABILITY. *Journal of Environmental Protection and Ecology*. https://www.scopus.com/pages/publications/105025904679

64. Tsang, Y.P., Choy, K.L., Wu, C.H., (...), Lam, H.Y. (2019) Blockchain-Driven IoT for Food Traceability with an Integrated Consensus Mechanism. *IEEE Access*. https://doi.org/10.1109/ACCESS.2019.2940227

65. Betz, A., Buchli, J., Göbel, C., Müller, C. (2015) Food waste in the Swiss food service industry - Magnitude and potential for reduction. *Waste Management*. https://doi.org/10.1016/j.wasman.2014.09.015

66. Engström, R., Carlsson-Kanyama, A. (2004) Food losses in food service institutions Examples from Sweden. *Food Policy*. https://doi.org/10.1016/j.foodpol.2004.03.004

67. Silvennoinen, K., Heikkilä, L., Katajajuuri, J.-M., Reinikainen, A. (2015) Food waste volume and origin: Case studies in the Finnish food service sector. *Waste Management*. https://doi.org/10.1016/j.wasman.2015.09.010

68. Shadid, N., Ahmed, V., Bahroun, Z. (2025) A systematic review of data-driven approaches to food waste and loss management. *Cleaner Waste Systems*. https://doi.org/10.1016/j.clwas.2025.100448

69. Forcada, O., Sert, S., Soldevila-Lafon, V., Vidali, F. (2017) Surplus food redistribution: Best practices from European countries. *Foodsaving in Europe: At the Crossroad of Social Innovation*. https://doi.org/10.1007/978-3-319-56555-2_5

70. Islam Khan, M.R. (2025) Machine Learning-Based Demand Forecasting for Reducing Food Waste in Perishable Supply Chains. *2025 IEEE 7th International Conference on Computing, Communication and Automation, ICCCA 2025*. https://doi.org/10.1109/ICCCA66364.2025.11325710

71. Kim, S., Gu, N., Moon, J., (...), Lee, K. (2023) Comparative analysis of model performance for predicting the customer of cafeteria using unstructured data. *Communications for Statistical Applications and Methods*. https://doi.org/10.29220/CSAM.2023.30.5.485

72. Laurent, S.K., Iskandar, R.J., Tjen, J. (2025) From Breakfast to Prediction: Unveiling Nasi Kuning Sales Patterns with Machine Learning. *2025 15th International Conference on Information and Communication Technology and System: AI for the Now and Next: Delivering Solutions and Driving Vision, ICTS 2025*. https://doi.org/10.1109/ICTS67612.2025.11369513

73. Vikanksha, V., Kumar, A., Singh, J. (2024) Reducing Food Waste Through Predictive Analytics. *Revolutionizing Sustainable Food Production With Quantum Computing*. https://doi.org/10.4018/979-8-3373-3957-3_9

74. Alukapally, N.R., Finny, J., Raj, G. (2025) Restaurants Profit Maximization Using Machine Learning Algorithms. *Lecture Notes in Electrical Engineering*. https://doi.org/10.1007/978-981-96-9203-3_36

75. Singh, B., Kaunert, C. (2024) Revolutionizing Responsible Consumption and Production of Food In Hospitality Industry: Machine Learning and Deep Learning Approach Towards Zero Food Waste. *Sustainable Waste Management in the Tourism and Hospitality Sectors*. https://doi.org/10.4018/979-8-3693-6110-8.ch017

76. Singh, B., Kaunert, C. (2024) Salvaging responsible consumption and production of food in the Hospitality Industry: Harnessing machine learning and deep learning for zero food waste. *Sustainable Disposal Methods of Food Wastes in Hospitality Operations*. https://doi.org/10.4018/979-8-3693-2181-2.ch012

77. Makin, A.H., Abdullah, M.I., Helmi, R.A. (2024) Demand Prediction for Perishable Products Using Machine Learning Regression Techniques: A Narrative Review. *2024 IEEE International Conference on Agrosystem Engineering, Technology and Applications: Integrating Smart Farming and Food Security for a Sustainable Future, AGRETA 2024*. https://doi.org/10.1109/AGRETA61912.2024.10948994

78. Zou, Q., Bezuidenhout, C.N., Ishrat, I. (2024) An application of machine learning to classify food waste interventions from a food supply chain perspective. *British Food Journal*. https://doi.org/10.1108/BFJ-02-2024-0135

79. Chae, B.K., Sheu, C., Park, E.O. (2024) The value of data, machine learning, and deep learning in restaurant demand forecasting: Insights and lessons learned from a large restaurant chain. *Decision Support Systems*. https://doi.org/10.1016/j.dss.2024.114291

80. Yoon, Gunwoo, Li, Cong, Choi, John Juyoung (2022) In search of time to bring the message on social media: Effects of temporal targeting and weather on digital consumers. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2022.1042714

81. Olawale, R.A., Olawumi, M.A., Oladapo, B.I. (2025) Sustainable farming with machine learning solutions for minimizing food waste. *Journal of Stored Products Research*. https://doi.org/10.1016/j.jspr.2025.102611

82. Ramanathan, R., Duan, Y., Ajmal, T., (...), Ramanathan, U. (2023) Motivations and Challenges for Food Companies in Using IoT Sensors for Reducing Food Waste: Some Insights and a Road Map for the Future. *Sustainability (Switzerland)*. https://doi.org/10.3390/su15021665

83. Sert, S., Garrone, P., Melacini, M., Perego, A. (2016) Surplus food redistribution for social purposes: The case of Coop Lombardia. *Organizing for Sustainable Effectiveness*. https://doi.org/10.1108/S2045-060520160000005015

84. Sundin, N., Citro, E., Eriksson, M. (2025) From waste to value? Balancing climate and social outcomes in social supermarkets. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2025.146950

85. Chakraborty, S., Sharma, A., Raj, S., Singh, N. (2025) A Smart Waste Food Management and Donation Platform Leveraging Machine Learning and Web Technologies. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-96-7134-2_39

86. Mustafa, M.F.M.S., Navaranjan, N., Demirovic, A. (2024) Food cold chain logistics and management: A review of current development and emerging trends. *Journal of Agriculture and Food Research*. https://doi.org/10.1016/j.jafr.2024.101343

87. Burgess, S., Wang, X., Rahbari, A., Hangi, M. (2022) Optimisation of a portable phase-change material (PCM) storage system for emerging cold-chain delivery applications. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2022.104855

88. Sergi, Ilaria, Montanaro, Teodoro, Benvenuto, Fabrizio Luca, Patrono, Luigi (2021) A Smart and Secure Logistics System Based on IoT and Cloud Technologies. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s21062231

89. Qian, J., Ruiz-García, L., Fan, B., (...), Wu, W. (2020) Food traceability system from governmental, corporate, and consumer perspectives in the European Union and China: A comparative review. *Trends in Food Science and Technology*. https://doi.org/10.1016/j.tifs.2020.03.025

90. Zuo, J., Feng, J., Gameiro, M.G., (...), He, Q. (2022) RFID-based sensing in smart packaging for food applications: A review. *Future Foods*. https://doi.org/10.1016/j.fufo.2022.100198

91. Shegar, S., Auti, A., Pathan, M., Pathan, T. (2026) A Next-Gen Intelligent System For Dairy Product Organicity And Traceability. *2026 2nd International Conference on Computing, Communication and Green Engineering, CCGE 2026*. https://doi.org/10.1109/CCGE67142.2026.11581867

92. Malik, M., Mor, R.S., Gahlawat, V.K., Kumar, V. (2026) Unlocking the potential: hybrid blockchain and AI-enabled traceability model development and implementation in the dairy industry – proof-of-concept. *Transportation Research Part E: Logistics and Transportation Review*. https://doi.org/10.1016/j.tre.2025.104552

93. Souprayen, S., Ayyasamy, A., Suresh Joseph, K.S. (2022) IoT-Blockchain driven traceability techniques for improved safety measures in food supply chain. *International Journal of Information Technology (Singapore)*. https://doi.org/10.1007/s41870-020-00581-y

94. Onyeaka, H., Siyanbola, K.F., Simon, Z. (2026) Blockchain and artificial intelligence in agriculture: Building resilient and sustainable food systems. *Urban Agriculture and Regional Food Systems*. https://doi.org/10.1002/uar2.70043

95. Huang, Y., Li, X., Chen, Y., (...), Adiljiang, N. (2025) Smart technologies for quality control and traceability in agri-food chains: A systematic review. *Proceedings of SPIE - The International Society for Optical Engineering*. https://doi.org/10.1117/12.3089999

96. Kılınç, I., Kilinc, B., Gevrekçı, Y., Takma, Ç. (2025) Enhancement of Hygiene and Safety in the Food Industry Through Digitalization With Blockchain Technology. *Journal of Food Safety and Food Quality*. https://doi.org/10.31083/JFSFQ42885

97. van Hilten, M., Ongena, G., Ravesteijn, P. (2020) Blockchain for Organic Food Traceability: Case Studies on Drivers and Challenges. *Frontiers in Blockchain*. https://doi.org/10.3389/fbloc.2020.567175

98. Hassoun, A., Jagtap, S., Garcia-Garcia, G., (...), Câmara, J.S. (2023) Food quality 4.0: From traditional approaches to digitalized automated analysis. *Journal of Food Engineering*. https://doi.org/10.1016/j.jfoodeng.2022.111216

99. Yu, P., Hamid, R., Hakim Osman, L., Liao, J. (2025) Bridging the digital gap: Empirical insights into agri-food supply chain transformation. *Sustainable Futures*. https://doi.org/10.1016/j.sftr.2025.100810

100. Wang, W., Yang, Z., Cao, Y., (...), Delen, D. (2025) An enhanced T-spherical fuzzy generalized multi-criteria decisioning model for evaluating the effectiveness of digital transformation in the sustainability of agri-food systems. *Engineering Applications of Artificial Intelligence*. https://doi.org/10.1016/j.engappai.2025.110887

101. Yusuf, H., Fors, H., Galal, N.M., (...), Harraz, N. (2025) Barriers to implementing circular citrus supply chains: A systematic literature review. *Journal of Environmental Management*. https://doi.org/10.1016/j.jenvman.2024.123963

102. Francksen, Sabrina, Ghaziani, Shahin, Bahrs, Enno (2025) Digital Maturity of Administration Entities in a State-Led Food Certification System Using the Example of Baden-Württemberg. *Foods*. https://doi.org/10.3390/foods14111870

103. Yang, Z., Wang, W.Z., Chen, Y., (...), Deveci, M. (2026) Analyzing barriers to AI-HI integration for food loss and waste reduction: A multi-actor study of food supply chains. *International Journal of Production Economics*. https://doi.org/10.1016/j.ijpe.2026.110146

104. Pradeepa, S.V., Derashri, P.D., Chowdhury, A.R., Tripathi, M.A. (2025) A Study on Food Supply Chain Management, System Implementation, and Future Research. *Essentials of Supply Chain Management: Strategy, Planning, and Operations*. https://www.scopus.com/pages/publications/105005052146

105. Solak, S., Scherrer, C., Ghoniem, A. (2014) The stop-and-drop problem in nonprofit food distribution networks. *Annals of Operations Research*. https://doi.org/10.1007/s10479-012-1068-7

106. Gunes, C., Van Hoeve, W.-J., Tayur, S. (2010) Vehicle routing for food rescue programs: A comparison of different approaches. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-642-13520-0_21